

THE FIFTEEN SPORADIC APÉRY-LIKE PAIRS: SECOND SOLUTIONS, THE χ -TWISTED LUCAS LAW, AND THE CLOSED FORMS

ABSTRACT. For each of the fifteen sporadic Apéry-like sequences $A(n)$ — Zagier’s six, the six Almkvist–Zudilin cases and Cooper’s three — let $B(n)$ be the second solution of the same three-term recurrence, normalised by $B(0) = 0$, $B(1) = 1$. The first row obeys a Lucas congruence $A(ap+r) \equiv A(a)A(r) \pmod{p}$. We report the current state of the second row. Exact computation across all fifteen pairs and all primes $5 \leq p \leq 103$ says that the naive analogue $p^w B(ap+r) \equiv B(a)A(r)$ is *false* for exactly seven of the fifteen, at exactly the primes at which a quadratic Dirichlet character χ takes the value -1 , and that inserting the single factor $\chi(p)$ repairs it with p -adic floor exactly w in every cell tested. We isolate the mechanism (one line of Frobenius descent for a χ -twisted harmonic letter), prove the descent under five explicit hypotheses, and correct the product-region hypothesis of the earlier statement: the surviving digit set is a proper subset and carries a base-level defect $\Delta(a)$, whose vanishing is an additional clause. The clause is automatic for tame weights, fails for α (Domb), ε and $\mathbf{E}$, and *no admissible weight repairs it* — for $\mathbf{E}$, the only known instance with $\chi \neq 1$, the defect is independent of the choice of decomposition. Four instances are theorems (γ , $\mathbf{A}$, $\mathbf{D}$, s_{10}), one (s_7) is one bounded local lemma away, three are blocked as above, and seven remain conjectural. Separately, a step-three endpoint transform proves the previously computational denominator bound $d_n^2 B(n) \in \mathbb{Z}$ for Zagier’s $\mathbf{B}$, shows the exponent two is optimal, and classifies its full dilation family by the exact condition $3 \mid h$. On the closed-form side we record a proof, formalised in Lean 4, of $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2 (2H_n^{(3)} - H_k^{(3)})$ for Apéry’s $\zeta(3)$ numerators; finite-certificate proofs, valid for every n , of the symmetric closed forms for the Franel numbers and for $\sum_k \binom{n}{k}^4$ — these sums are the t^2 -coefficients of Straub–Zudilin’s deformation, and we say so, together with an elementary re-derivation of the Apéry limits $\zeta(2)/4$ and $\zeta(2)/5$ that they proved for all exponents; a proof that the mechanism behind those two *cannot* close the $\zeta(2)$ -Apéry case; and a χ_{-4} -twisted closed form for the Catalan-constant family $\mathbf{E}$. Three of the fifteen have no archimedean Apéry limit at all and satisfy the p -adic law regardless: the descent is a statement about Frobenius. We locate the law precisely against Beukers–Vlasenko’s Conjecture 7.5: the two are the two rows of one 2×2 Frobenius matrix, neither implies the other, and the character is invisible in unit-root formulations.

1. INTRODUCTION

1.1. The two rows. Apéry’s proof that $\zeta(3) \notin \mathbb{Q}$ rests on a pair of solutions of one three-term recurrence: the integral solution $a_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ and a companion b_n with $b_n/a_n \rightarrow \zeta(3)$. The arithmetic of the two rows is not symmetric. The first row is integral and satisfies a Lucas congruence,

$$A(ap+r) \equiv A(a)A(r) \pmod{p} \quad (0 \leq a, r < p), \quad (1.1)$$

proved for a_n by Gessel [9], extended to the sporadic class by Malik–Straub [11], obtained for fourteen of the fifteen in a strong form by Gorodetsky [10], and completed for all fifteen by Straub [16], who resolves the two remaining cases with constant-term representations. The second row is not integral — it has denominators d_n^w , $d_n = \text{lcm}(1, \dots, n)$ — and no congruence of Lucas type for it appears in the literature.

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This paper is a status report on the second row for the whole sporadic class. It collects what is proved, states honestly what is only computed, and — this is its main technical purpose — records two corrections to the companion papers [21, 22], and assembles the closed forms that the congruence machinery consumes.

1.2. The law. Write $q = \lfloor n/p \rfloor$. For each sporadic pair there is an integer $w \in \{2, 3\}$ and a quadratic Dirichlet character χ (possibly trivial) such that, empirically,

$$v_p(p^w B(n)A(q) - \chi(p)B(q)A(n)) \geq w, \quad (\text{LB}_w^\chi)$$

with equality in every cell tested. The untwisted form ($\chi \equiv 1$) is *false* for seven of the fifteen families, and the failing primes are exactly those with $\chi(p) = -1$; there the valuation drops to 0 in the single-digit range and as low as the integrality bound permits beyond it. The character is the character of the sequence's Apéry limit: the limit is a rational multiple of $\zeta(w)$ when χ is trivial and of $L(\chi, w)$ otherwise.

We prove the single-digit, mod- p shadow of (LB_w^χ) ,

$$p^w B(ap + r) \equiv \chi(p)^e B(a) A(r) \pmod{p} \quad (1 \leq a < p, 0 \leq r < p), \quad (1.2)$$

under hypotheses (H1)–(H5) of §4, and verify those hypotheses for four of the fifteen. The full statement (LB_w^χ) — all n , modulus p^w — is [VERIFIED] only and is stated as Conjecture 3.1; we never call it a theorem.

1.3. What is new here.

- (1) **A correction to the product-region hypothesis** (§4). The hypothesis (H2) used in [21] asserts that the set of surviving base- p digits is $\{0 \leq b \leq a\} \times \Sigma_r$. That is false: the b -side is a proper subset for every $a \geq (p+1)/2$, already for the Apéry $\zeta(3)$ pair, and the omitted terms contribute a *base-level defect* $\Delta(a)$ whose vanishing mod p is a further clause. The clause is automatic under tameness, so the four proved instances are unaffected; it is exactly what fails for $\alpha, \varepsilon, \mathbf{E}$.
- (2) **A proved negative** (§5.4). No admissible harmonic weight repairs $\Delta(a)$ for α, ε or $\mathbf{E}$; for $\mathbf{E}$ the class $\Delta(a) \pmod{p}$ is an invariant of the family, not of the fitted weight. Consequently the earlier plan — “prove one valuation inequality and move four families inside the theorem” — is wrong; it works for s_7 alone.
- (3) **Two closed forms with complete finite certificates** (§6). Symmetric harmonic weights for the Franel numbers and for $\sum_k \binom{n}{k}^4$, with every certificate identity re-verified symbolically here, valid for every $n \geq 1$. These sums are *not* new: they are the t^2 -coefficients of Straub–Zudilin’s analytic deformation [17], as we show in Remark 6.8, and their Apéry limits $\zeta(2)/(d+1)$ are theorems of *loc. cit.* What is contributed here is the finite certificate proof — Straub–Zudilin obtain the recurrence only for large enough n , and their route passes through a locally uniform limit — and the explicit harmonic-monomial display, which is what Theorem 4.5 consumes.
- (4) **A provable insufficiency** (§6.4). The mechanism that proves those two closed forms provably fails for the $\zeta(2)$ -Apéry pair: one component of the correction is not Gosper-summable. This is the first place in the programme where the rank-two mechanism is *provably* insufficient.

1.4. Evidence classes. The programme’s labelling discipline is retained verbatim in this paper, and the labels are load-bearing. [THM]: proved and adversarially refereed. [PROVED]: a complete proof is given or cited to a written proof. [LEAN]: checked by the Lean 4 kernel with no `sorry`. [VERIFIED] (with a range): an exact finite computation, *never* treated as proof. [EXCLUDED]: a proved negative. [OPEN]. A statement carrying [VERIFIED] is never restated later as though it carried [PROVED].

1.5. Relation to the companion papers. The weight-three congruence (1.2) for the Apéry $\zeta(3)$ pair, its elementary self-contained proof, the sharpness analysis at $p = 2, 3$ and the general theorem are the subject of [21]; the closed form for b_n together with its Lean formalisation is [22]. We do not repeat those proofs. What is here is the class-wide picture, the corrections of items 1–2 above, and the closed-form material of §6.

1.6. Conventions. p is a prime, $\mathbb{Z}_{(p)}$ the localisation of $\mathbb{Z}$ at p , v_p the p -adic valuation on $\mathbb{Q}$, and $x \equiv y \pmod{p^m}$ for $x, y \in \mathbb{Q}$ means $v_p(x - y) \geq m$. We write $H_y^{(r)} = \sum_{m=1}^y m^{-r}$ and, for a Dirichlet character χ , $K_\chi^{(r)}(y) = \sum_{m=1}^y \chi(m)m^{-r}$; thus $H_y^{(r)} = K_1^{(r)}(y)$. χ_D is the Kronecker symbol $(\frac{D}{\cdot})$, and $L(\chi, s) = \sum_{m \geq 1} \chi(m)m^{-s}$.

2. THE FIFTEEN PAIRS

2.1. The recurrences. Two families of three-term recurrences, with $u_{-1} = 0$, $u_0 = 1$:

$$(n+1)^2 u_{n+1} = (an^2 + an + b)u_n - cn^2 u_{n-1}, \quad (2.1)$$

$$(n+1)^3 u_{n+1} = (2n+1)(an^2 + an + b)u_n - n(cn^2 + d)u_{n-1}. \quad (2.2)$$

Up to degenerate cases, (2.1) has six sporadic integral solutions, found by Zagier [20] and labelled **A**, $\dots$, **F**; (2.2) with $d = 0$ has six, found by Almkvist–Zudilin [1] and labelled $\alpha, \gamma, \delta, \varepsilon, \zeta, \eta$; and Cooper [7] found three further solutions with $d \neq 0$ (so counted in [11, §2]; [7] itself announces two), labelled s_7, s_{10}, s_{18} after the levels of the associated modular forms; they satisfy two-term supercongruences [13], and Cooper has since extended the search to four-term recurrences [8]. We follow the labelling and the binomial sums of Malik–Straub [11, Tables 1–2], whose second column records the alternative Almkvist–van Straten–Zudilin labels (a), $\dots$, (g).

Definition 2.1. For each pair, $B(n)$ is the second solution: $B(0) = 0$, $B(1) = 1$, and the recurrence is imposed for $n \geq 1$ (the step $n = 0$ is degenerate). This is Apéry’s own normalisation and that of Chamberland–Straub [6]. Rescaling B by a p -unit rational does not affect any statement below, so for $p \geq 5$ the normalisation is harmless.

2.2. The table. Table 1 lists the fifteen pairs with their parameters, first row, Apéry limit $L = \lim B(n)/A(n)$, weight w and character χ .

Here $s_{18}(0) = 1$ and, for $n \geq 1$,

$$s_{18}(n) = \sum_{k=0}^{\lfloor n/3 \rfloor} (-1)^k \binom{n}{k} \binom{2k}{k} \binom{2(n-k)}{n-k} \left[\binom{2n-3k-1}{n} + \binom{2n-3k}{n} \right]. \quad (2.3)$$

[VERIFIED: $n \leq 12$, exact] each binomial sum in Table 1 satisfies its recurrence, re-checked independently for this paper.

2.3. The limits. [VERIFIED: 400 digits] The limits in Table 1 were identified by two-term integer relation detection against a 33-constant basis ($\zeta(2), \zeta(3), \zeta(4)$; $L(\chi_D, 2)$, $L(\chi_D, 3)$ for $D = -3, -4, -7, -8, -11, -15, -20, -24, 5, 8, 12, 13, 24$; $\log^2 2$, $\pi^2 \log 2$, $\log^3 2$), matching to between 31 and 401 digits (the slowest, **F**, converges at rate $(8/9)^n$). *We claim no novelty for any of these values.* Almkvist–van Straten–Zudilin [2, §2] tabulate the limits of the six weight-two families — $\pi^2/24$ (**A**), $\pi^2/30$ (**D**), $L(\chi_{-3}, 2)/2$ (**C**), $G/2$ (**E**), $\frac{5}{8}L(\chi_{-3}, 2)$ (**F**) — and record that the remaining one, **B**, is *not convergent*, attributing the proofs to Zagier [20] and Yang [19]. The Franel and $\sum_k \binom{n}{k}^4$ limits are also [6, eq. (20)] (Cusick, via [18, p. 202]) and are proved in general by Straub–Zudilin [17]; $\zeta(3)/6$ is Apéry’s; and the Domb limit $7\zeta(3)/24$, for exactly the normalisation of Definition 2.1, is proved by Shvets [15]. The values for $\varepsilon, \zeta, s_7, s_{18}$ and the non-convergence of δ and η we have not located in the literature, but they are of the shape predicted by [20, 1] and we record them only as [VERIFIED].

#	family	$(a, b, c[, d])$	$A(n)$	L	w	χ
1	A (a)	(7, 2, -8)	$\sum_k \binom{n}{k}^3$ (Franel)	$\zeta(2)/4$	2	1
2	B (f)	(9, 3, 27)	$\sum_k (-1)^k 3^{n-3k} \binom{n}{3k} \frac{(3k)!}{k!^3}$	none	2	χ_{-3}
3	C (c)	(10, 3, 9)	$\sum_k \binom{n}{k}^2 \binom{2k}{k}$	$L(\chi_{-3}, 2)/2$	2	χ_{-3}
4	D (b)	(11, 3, -1)	$\sum_k \binom{n}{k}^2 \binom{n+k}{n-k}$	$\zeta(2)/5$	2	1
5	E (d)	(12, 4, 32)	$\sum_k \binom{n}{k}^2 \binom{2k}{k} \binom{2(n-k)}{n-k}$	$G/2$	2	χ_{-4}
6	F (g)	(17, 6, 72)	$\sum_k (-1)^k 8^{n-k} \binom{n}{k} \sum_l \binom{k}{l}^3$	$\frac{5}{8} L(\chi_{-3}, 2)$	2	χ_{-3}
7	α	(10, 4, 64, 0)	$\sum_k \binom{n}{k}^2 \binom{2k}{k} \binom{2(n-k)}{n-k}$ (Domb)	$7\zeta(3)/24$	3	1
8	γ	(17, 5, 1, 0)	$\sum_k \binom{n}{k}^2 \binom{n+k}{n}^2$ (Apéry)	$\zeta(3)/6$	3	1
9	δ	(7, 3, 81, 0)	$\sum_k (-1)^k 3^{n-3k} \binom{n}{3k} \binom{n+k}{n} \frac{(3k)!}{k!^3}$	none	3	1
10	ε	(12, 4, 16, 0)	$\sum_k \binom{n}{k}^2 \binom{2k}{k}^2$	$7\zeta(3)/32$	3	1
11	ζ	(9, 3, -27, 0)	$\sum_{k,l} \binom{n}{k}^2 \binom{n}{l} \binom{k}{l} \binom{k+l}{n}$	$L(\chi_{-3}, 3)/3$	3	χ_{-3}
12	η	(11, 5, 125, 0)	$\sum_k (-1)^k \binom{n}{k}^3 \left[\binom{4n-5k-1}{3n} + \binom{4n-5k}{3n} \right]$	none	3	χ_5
13	s_7	(13, 4, -27, 3)	$\sum_k \binom{n}{k}^2 \binom{n+k}{k} \binom{2k}{n}$	$\zeta(2)/7$	2	1
14	s_{10}	(6, 2, -64, 4)	$\sum_k \binom{n}{k}^4$	$\zeta(2)/5$	2	1
15	s_{18}	(14, 6, 192, -12)	see (2.3)	$L(\chi_{-3}, 2)/2$	2	χ_{-3}

TABLE 1. The fifteen sporadic pairs. Parenthesised letters in column 2 are the Almkvist–van Straten–Zudilin labels of [11, Table 1]. $G = L(\chi_{-4}, 2)$ is Catalan’s constant. Rows 1–6 satisfy (2.1); rows 7–12 satisfy (2.2) with $d = 0$; rows 13–15 satisfy (2.2) with $d \neq 0$.

2.4. The weights, and their sharpness.

Observation 2.2. [VERIFIED: $n \leq 600$; independently $n \leq 120$ for this paper] Let w be the least exponent with $d_n^w B(n) \in \mathbb{Z}$ for all n in the tested range, $d_n = \text{lcm}(1, \dots, n)$. Then

$$\begin{aligned} w = 2 & \quad \text{for all six of (2.1) and all three of Cooper’s } d \neq 0 \text{ cases,} \\ w = 3 & \quad \text{for all six Almkvist–Zudilin } d = 0 \text{ cases.} \end{aligned}$$

It is sharp: $d_n^{w-1} B(n) \notin \mathbb{Z}$ for at least 596 of the 600 values of n in the first sweep (and for 116–119 of the 120 in the second). Per prime, $v_p(B(n)) \geq -w \lfloor \log_p n \rfloor$, [VERIFIED: 0 failures, 15 families $\times$ 16 primes $\times$ $n \leq 400$].

Two things are worth isolating. First, w tracks the Cooper deformation $d \neq 0$, not the shape of the recurrence: Cooper’s three live in the cubic family (2.2) and have weight 2, not 3. Second, wherever the Apéry limit exists it is a rational multiple of $\zeta(w)$ or $L(\chi, w)$, so w agrees with the motivic weight of the limit — 12 cases out of 12.

2.5. Three pairs with no archimedean limit.

Proposition 2.3. [PROVED] *For **B**, δ and η the characteristic equation of the recurrence has a pair of complex-conjugate roots of equal modulus, with discriminants*

$$\mathbf{B} : a^2 - 4c = -27, \quad \delta : 4a^2 - 4c = -128, \quad \eta : 4a^2 - 4c = -16,$$

so that Poincaré–Perron provides no dominant solution and $B(n)/A(n)$ does not converge. Every other sporadic pair has real distinct characteristic roots.

Proof. The characteristic equation of (2.1) is $\lambda^2 - a\lambda + c = 0$ and of (2.2) is $\lambda^2 - 2a\lambda + c = 0$; substitute the parameters of Table 1. The discriminants of the remaining twelve are 81, 64, 125, 16, 1, 144, 1152, 512, 432, 784, 400, 16, all positive. $\square$

[VERIFIED: $n = 590$ vs. $n = 600$] the ratios $B(n)/A(n)$ for these three agree to 0 digits, against 380 digits for the convergent families. Nevertheless all three satisfy (LB_w^χ) flat, with $\chi = \chi_{-3}, \mathbf{1}, \chi_5$ respectively. This is the cleanest evidence in the programme that (LB_w^χ) is a statement about *Frobenius*, not about the archimedean limit; we return to it in §7.

3. THE χ -TWISTED LAW

3.1. Where the untwisted law dies. [VERIFIED: $5 \leq p \leq 43, n \leq 300$, exact rational arithmetic] Testing $v_p(p^w B(n)A(q) - B(q)A(n)) \geq w$ with $q = \lfloor n/p \rfloor$:

verdict	families
holds flat at every prime tested	$\mathbf{A}, \mathbf{D}, \alpha, \gamma, \delta, \varepsilon, s_7, s_{10}$ (8)
fails at $p \equiv 2 \pmod{3}$	$\mathbf{B}, \mathbf{C}, \mathbf{F}, \zeta, s_{18}$ (5)
fails at $p \equiv 3 \pmod{4}$	$\mathbf{E}$ (1)
fails at $p \equiv \pm 2 \pmod{5}$	η (1)

The failures are total, not marginal: at a failing prime the single-digit floor is 0 (the difference is a p -unit) and in the multi-digit range the valuation falls to $-w \cdot (\# \text{digits} - 1)$, i.e. as low as Observation 2.2 permits. They are not explained by exceptional digits ($p \mid A(a)$), by the centre residue, or by $v_p\binom{2n}{n}$: the failing set is exactly $\{p : \chi(p) = -1\}$ for one quadratic χ . The cross-checks that isolate the character are [VERIFIED]: the passing primes of the χ_{-3} group are mixed mod 4 and mod 5; those of $\mathbf{E}$ are mixed mod 3 and mod 5; and those of η in the tested range, 11, 19, 29, 31, 41, are mixed mod 3 and mod 4 and are precisely the $p \equiv \pm 1 \pmod{5}$.

3.2. The repair.

Conjecture 3.1 (master form). *For every sporadic pair, with w and χ as in Table 1, every prime $p \geq 5$ and every $n \geq 1$, writing $q = \lfloor n/p \rfloor$,*

$$v_p(p^w B(n)A(q) - \chi(p)B(q)A(n)) \geq w.$$

[VERIFIED: 0 failures] all 15 pairs, all primes $5 \leq p \leq 103, n \leq 400$; and $n \leq 1200$ for $p = 5, 7, 11$ — five base- p digits at $p = 5$. In every one of the $15 \times 16 = 240$ (family, prime) cells with $5 \leq p \leq 61$ the floor is *exactly* w : perfectly flat, with no dependence on the digit pattern, no centre-residue exception and no exceptional-digit exception. Independently for this paper: all 15 families at $p = 5$ and $p = 7$, all $n < p^3$, floor exactly w in all 30 cells.

Remark 3.2 (the ramified prime). For η the prime $p = 5$ is ramified, $\chi_5(5) = 0$, and Conjecture 3.1 degenerates to $v_p(p^3 B(n)A(q)) \geq 3$, i.e. to the 5-integrality of $B(n)$. That is true: [VERIFIED: $n \leq 500$] $\min_n v_5(B(n)) = 0$, the pole simply does not develop at the ramified prime, and the floor is exactly 3. This is the only prime ≥ 5 at which any of the fifteen characters ramifies; the χ_{-3} and χ_{-4} families ramify only at 3 and 2.

Remark 3.3 ($p = 2, 3$). Excluded, correctly. With the twisted law and $n \leq 120$ the floor drops below w for $\mathbf{A}$ ($p = 2$, floor 1), $\mathbf{F}$ ($p = 2$, 1), α ($p = 2$ and $p = 3$, 2), γ ($p = 3$, 2), δ ($p = 3$, 1), ε ($p = 2$, 2). See [21, §8] for the analysis of where $p \geq 5$ enters.

4. LEMMA K AND THEOREM LB

4.1. The source of the character.

Lemma 4.1 (Lemma K). [PROVED] *Let χ be completely multiplicative (a Dirichlet character; $\chi \equiv 1$ allowed), let $r \geq 1$ and $y \geq 0$. Then, as an identity in $\mathbb{Q}$,*

$$p^r K_\chi^{(r)}(y) = \chi(p) K_\chi^{(r)}(\lfloor y/p \rfloor) + p^r T, \quad T := \sum_{m \leq y, p \nmid m} \frac{\chi(m)}{m^r} \in \mathbb{Z}_{(p)}.$$

Proof. Split the sum according to $p \mid m$. The p -divisible part is $\sum_{j \leq \lfloor y/p \rfloor} \chi(jp)(jp)^{-r} = \chi(p)p^{-r} \sum_{j \leq \lfloor y/p \rfloor} \chi(j)j^{-r}$, by complete multiplicativity. The complementary part has p -integral terms. $\square$

This one line is the entire source of the factor $\chi(p)$ in (1.2): a second solution whose weight needs χ -twisted harmonic letters inherits $\chi(p)$ when the letters descend, and one whose weight is a pure ζ -harmonic monomial inherits $+1$.

Corollary 4.2. [PROVED] *Let $M = \prod_{t=1}^s K_{\chi_t}^{(r_t)}(x_t)$ with $\sum_t r_t = w$, and suppose every $K_{\chi_t}^{(r_t)}(\lfloor x_t/p \rfloor)$ lies in $\mathbb{Z}_{(p)}$. Then $p^w M \equiv (\prod_t \chi_t(p)) \prod_t K_{\chi_t}^{(r_t)}(\lfloor x_t/p \rfloor) \pmod{p}$.*

Proof. $p^w M = \prod_t (\chi_t(p) K_{\chi_t}^{(r_t)}(\lfloor x_t/p \rfloor) + p^{r_t} T_t)$ by Lemma 4.1; expand. Every factor is p -integral and every cross term carries a factor p . $\square$

4.2. The hypotheses, corrected. Fix a prime p and write $n = ap + r$, $k = bp + s$ with $1 \leq a < p$, $0 \leq r, s < p$. Let

$$S(n, k) = \prod_i \binom{L_i(n, k)}{M_i(n, k)}, \quad A(n) = \sum_k S(n, k), \quad B(n) = \sum_k S(n, k) \mathfrak{w}(n, k),$$

with L_i, M_i integer linear forms, and let the weight be a $\mathbb{Q}$ -combination of monomials of total degree w ,

$$\mathfrak{w}(n, k) = \sum_j c_j \prod_t K_{\chi}^{(r_{jt})}(x_{jt}(n, k)), \quad \sum_t r_{jt} = w, \quad (4.1)$$

in letters attached to one fixed quadratic character χ and to the trivial character, with x_{jt} integer linear forms. Put $\mathcal{B}_a = \{b : p \nmid S(a, b)\}$ and define the *base-level defect*

$$\Delta(a) := \sum_{b \notin \mathcal{B}_a} S(a, b) \mathfrak{w}(a, b) = B(a) - \sum_{b \in \mathcal{B}_a} S(a, b) \mathfrak{w}(a, b). \quad (4.2)$$

- (H1) *Lucas/carry dichotomy.* For every k , either $p \mid S(n, k)$ or $S(n, k) \equiv S(a, b)S(r, s) \pmod{p}$.
- (H2) *Product region and base-level defect.* The surviving set $\{k : p \nmid S(n, k)\}$ is $\mathcal{B}_a \times \Sigma_r$ for some $\Sigma_r \subseteq [0, p)$; $\sum_{s \in \Sigma_r} S(r, s) \equiv A(r) \pmod{p}$; and $\Delta(a) \equiv 0 \pmod{p}$.
- (H3) *Digit compatibility.* On the surviving set, $\lfloor x(n, k)/p \rfloor = x(a, b)$ for every argument form x occurring in $\mathfrak{w}$.
- (H4) *Tameness.* $0 \leq x(n, k) \leq n$ on the summation range for every argument form x ; and $c_j \in \mathbb{Z}_{(p)}$ for every j .
- (H5) *χ -homogeneity.* Every monomial of $\mathfrak{w}$ contains the same number $e \in \{0, 1\}$ of χ -letters, the others being trivial-character letters.

Remark 4.3 (what was corrected, and why it matters). The version of (H2) used in [21] reads “the surviving set is $\{0 \leq b \leq a\} \times \Sigma_r$ ”, with no Δ -clause. The b -side is *not* $\{0, \dots, a\}$ in general. Already for γ , with $S(n, k) = \binom{n}{k}^2 \binom{n+k}{n}^2$, one has $p \mid \binom{a+b}{a}$ as soon as $a + b \geq p$, so digits b with $b > p - 1 - a$ drop out, and the b -side is proper for every $a \geq (p+1)/2$; the same happens for **D** and, with a different mechanism, for α (at $p = 5$, $a = 3$ the digit $b = 0$ is dropped because $5 \mid \binom{6}{3}$). [VERIFIED: all four non-tame families, $5 \leq p \leq 23$] The omitted terms are collected by $\Delta(a)$, and the proof below needs $\Delta(a) \equiv 0$. Lemma 4.4 shows the clause is free under (H4), so the conclusions of [21] for the four tame families stand as stated; the clause is nevertheless the exact obstruction for α , ε and **E** (§5.4).

Lemma 4.4. [PROVED] *If (H4) holds then $\mathfrak{w}(a, b) \in \mathbb{Z}_{(p)}$ for all b , and hence $\Delta(a) \equiv 0 \pmod{p}$ automatically.*

Proof. By (H4), every argument satisfies $x(a, b) \leq a < p$, so no denominator of $K_{\chi}^{(r)}(x(a, b))$ is divisible by p ; with $c_j \in \mathbb{Z}_{(p)}$ this gives $\mathfrak{w}(a, b) \in \mathbb{Z}_{(p)}$. Each term of (4.2) then has $p \mid S(a, b)$ times a p -integral factor. $\square$

4.3. The theorem.

Theorem 4.5 (Theorem LB). [PROVED] *Under (H1)–(H5), for all $1 \leq a < p$ and $0 \leq r < p$,*

$$p^w B(ap + r) \equiv \chi(p)^e B(a) A(r) \pmod{p}.$$

Proof. Let $n = ap + r$. By (H4) every argument satisfies $x(n, k) \leq n < p^2$, so $\lfloor x(n, k)/p \rfloor \leq a < p$ and each descended letter $K_X^{(r)}(\lfloor x(n, k)/p \rfloor)$ lies in $\mathbb{Z}_{(p)}$; likewise $x(a, b) \leq a < p$, so $\mathfrak{w}(a, b) \in \mathbb{Z}_{(p)}$, and $p^w \mathfrak{w}(n, k) \in \mathbb{Z}_{(p)}$ for every k . Corollary 4.2 and (H5) give, for every k ,

$$p^w \mathfrak{w}(n, k) \equiv \chi(p)^e \tilde{\mathfrak{w}}(n, k) \pmod{p}, \quad \tilde{\mathfrak{w}}(n, k) = \sum_j c_j \prod_t K_X^{(r_{jt})}(\lfloor x_{jt}(n, k)/p \rfloor),$$

and (H3) upgrades $\tilde{\mathfrak{w}}(n, k) = \mathfrak{w}(a, b)$ on the surviving set. Now split

$$p^w B(n) = \sum_{k: p|S(n, k)} S(n, k) p^w \mathfrak{w}(n, k) + \sum_{k \text{ surviving}} S(n, k) p^w \mathfrak{w}(n, k).$$

Every $p^w \mathfrak{w}(n, k)$ is p -integral, so the first sum is $\equiv 0 \pmod{p}$. In the second, substitute (H1) and the displayed congruence and use the product structure of (H2):

$$p^w B(n) \equiv \chi(p)^e \left(\sum_{b \in \mathcal{B}_a} S(a, b) \mathfrak{w}(a, b) \right) \left(\sum_{s \in \Sigma_r} S(r, s) \right) \equiv \chi(p)^e (B(a) - \Delta(a)) A(r) \pmod{p},$$

and the clause $\Delta(a) \equiv 0$ of (H2) finishes. $\square$

Remark 4.6 (the hypotheses in context). (H1) and the product structure in (H2) are exactly the Lucas-factorisation-plus-carry-annihilation property established for all fifteen summands by Malik–Straub [11], so they are available class-wide. What is *not* needed is a per-carry valuation bound: (H4) makes $p^w \mathfrak{w}$ p -integral outright. Where tameness fails one needs a substitute, and §5.4 shows what that substitute has to be.

Remark 4.7 (the formalised variant). The Lean formulation (§8) replaces (H1) and the product clause of (H2) by the single unconditional congruence $S(ap + r, bp + s) \equiv S(a, b) S(r, s) \pmod{p}$ for all a, b and all $r, s < p$, together with $S(n, k) = 0$ for $k > n$. In the carrying regime both sides vanish mod p , so the dichotomy is absorbed; the double sum then factors because a *full* block $[0, (a + 1)p)$ is split and both marginal sums are complete. The formalised theorem is therefore immune to the defect of Remark 4.3 — there is no product-region argument in it at all — at the cost of demanding the digit congruence for every b , not only for $b \leq a$. This is a genuine simplification and we recommend it as the definitive form of the hypothesis.

Observation 4.8 (integrality). [VERIFIED: 0 failures, 15 families $\times$ 16 primes $\times$ $n \leq 400$] $v_p(B(n)) \geq -w \lfloor \log_p n \rfloor$, i.e. $d_n^w B(n) \in \mathbb{Z}_{(p)}$.

Remark 4.9. This is *not* a corollary of Theorem 4.5. That theorem is a single-digit statement — (H4) is used in the form $x(n, k) \leq n < p^2$, which is precisely $a < p$ — and the iteration over base- p digits would need it at $a \geq p$, where the descended letters $K_X^{(r)}(\lfloor x/p \rfloor)$ are no longer p -integral. Even granting Conjecture 3.1, passing to the ratio form $p^w B(n)/A(n) \equiv \chi(p) B(q)/A(q)$ requires $A(n)$ and $A(q)$ to be p -units, which is a separate matter (see [11] for the primes that never divide a given sporadic sequence). Apart from the **B** case proved in Theorem 4.10 below, we therefore record Observation 4.8 as computed, and in §7 treat the implication from $(\text{LB}_w^\times)$ to mirror-map integrality as a heuristic and not as a proof.

Theorem 4.10 (sharp denominator for **B**). *For $h \in \mathbb{Z}$, let $B_h(0) = 0$, $B_h(1) = 1$, and*

$$(n + 1)^2 B_h(n + 1) = h(3n^2 + 3n + 1) B_h(n) - 3h^2 n^2 B_h(n - 1).$$

Then

$$(\forall n \geq 0, d_n^2 B_h(n) \in \mathbb{Z}) \iff 3 \mid h.$$

At $h = 3$ this is Zagier's sporadic pair **B**. Hence the **B** row of Observation 4.8 is a theorem, and its exponent 2 is optimal.

Proof. Set

$$T_h(n) = \sum_{k=0}^n \binom{n}{k} (-h)^{n-k} B_h(k), \quad T_h(m) = 0 \quad (m \leq 0).$$

Conjugating the recurrence by this binomial transform gives the step-three endpoint recurrence

$$n^2 T_h(n) + h^3(n-2)(n-1)T_h(n-3) = (-h)^{n-1}.$$

Indeed, at the generating-function level the original operator

$$\theta^2 - hz(3\theta^2 + 3\theta + 1) + 3h^2 z^2(\theta + 1)^2$$

is conjugated to

$$\theta^2 + h^3 z^3(\theta + 1)(\theta + 2).$$

Iteration yields

$$(-1)^{n-1} T_h(n) = h^{n-1} \sum_{\substack{0 \leq j < n \\ n-j \equiv 1 \pmod{3}}} \frac{\prod_{t=0}^{r-1} (j+2+3t)(j+3+3t)}{\prod_{t=0}^r (j+1+3t)^2}, \quad r = \frac{n-j-1}{3}. \quad (\text{B-end})$$

All propagation signs agree because $j+r \equiv n-1 \pmod{2}$.

Let $R(n, j)$ denote the quotient in (B-end). If $p \neq 3$, counting multiples of each p^a in the three arithmetic progressions shows

$$v_p(R(n, j)) \geq -2v_p(d_n) :$$

the doubled denominator progression has length $r+1$, the two numerator progressions have length r , and the deficit for each p^a is at most 2. For $p = 3$, choose a factor of maximal 3-adic valuation in $D = \prod_{t=0}^r (j+1+3t)$. The difference of any other factor from it is 3 times an index difference, whence

$$v_3(D) \leq v_3(d_n) + r + v_3(r!), \quad v_3(R(n, j)) \geq -2v_3(d_n) - 2r - 2v_3(r!).$$

If $3 \mid h$, then $n-1 = j+3r \geq 3r$ and $v_3(r!) \leq r/2$, so every summand of $d_n^2 T_h(n)$ in (B-end) is integral. Binomial inversion and $d_k \mid d_n$ transfer the result to $B_h(n)$.

Conversely, direct recurrence propagation gives

$$B_h(6) = -\frac{39521h^5}{32400}, \quad d_6^2 B_h(6) = -\frac{39521h^5}{9}.$$

Since $3 \nmid 39521$, integrality at $n = 6$ forces $3 \mid h$. Finally $B_3(2) = 21/4$, so $d_2 B_3(2) = 21/2 \notin \mathbb{Z}$; the exponent 2 is sharp. $\square$

5. STATUS OF THE FIFTEEN

Table 2 is the summary; the four subsections that follow justify it.

family	$\mathfrak{w}$ known?	tame?	$\Delta \equiv 0$?	status of (1.2)
γ	yes, 2 terms	yes	free	[THM] $p \geq 5$; [LEAN]
A	yes, 3 terms	yes	free	[THM] $p \geq 5$ ($p \neq 2$ suffices); [LEAN]
D	yes, 4 terms	yes	free	[THM] $p \geq 7$; $p = 5$ [VERIFIED] only
s_{10}	yes, 3 terms	yes	free	[THM] $p \geq 7$
s_7	yes, 8 terms	no	[VERIFIED] $p \leq 37$	reachable; two local bounds, [OPEN]
α	yes, 14 terms	no	[EXCLUDED]	blocked; needs (LD)
ε	yes, 10 terms	no	[EXCLUDED]	blocked; needs (LD)
E	yes, 5 terms, χ_{-4}	no	[EXCLUDED] (inv.)	blocked; needs (LD)
B, C, F, $\delta, \zeta, \eta, s_{18}$	no ([EXCLUDED])	—	—	conjectural

TABLE 2. Status of the second-row congruence (1.2). “Tame” means every letter argument lies in $[0, n]$ on the summation range, i.e. (H4). “ $\Delta \equiv 0$ free” means Lemma 4.4 applies.

5.1. The harmonic decompositions. All of the following were obtained by exact linear fitting of an ansatz $B(n) = \sum_k S(n, k) \mathfrak{w}(n, k)$ with $\mathfrak{w}$ as in (4.1), solved over a large prime field, rationally reconstructed, then *validated exactly over $\mathbb{Q}$ on held-out n* . Each row below is [VERIFIED: held out $n = 60, \dots, 72$, exact; and re-validated for this paper at $n \leq 20$].

family	w	$\mathfrak{w}(n, k)$	arguments
γ	3	$\frac{1}{3}H_n^{(3)} - \frac{1}{6}H_k^{(3)}$	n, k
A	2	$\frac{1}{4}H_k^{(2)} + \frac{3}{4}H_k^{(1)}(H_k^{(1)} - H_{n-k}^{(1)})$	$k, n - k$
D	2	$\frac{1}{5}[H_n^{(2)} + H_k^{(1)}(2H_k^{(1)} - H_{n-k}^{(1)} - H_n^{(1)})]$	$n, k, n - k$
s_{10}	2	$\frac{1}{5}H_k^{(2)} + \frac{4}{5}H_k^{(1)}(H_k^{(1)} - H_{n-k}^{(1)})$	$k, n - k$
s_7	2	8 terms in $H^{(1)}, H^{(2)}$	$k, n - k, n, 2k$
ε	3	10 terms in $H^{(1)}, H^{(2)}, H^{(3)}$	$k, n - k, 2k, 2k - n$
α	3	14 terms in $H^{(1)}, H^{(2)}, H^{(3)}$	$k, n - k, 2k, 2n - 2k$
E	2	$\frac{1}{2}K_{\chi_{-4}}^{(2)}(2k) + (\frac{3}{4}H_k^{(1)} - \frac{1}{2}H_{2k}^{(1)})(K_{\chi_{-4}}^{(1)}(2k) - K_{\chi_{-4}}^{(1)}(2n - 2k))$	$k, 2k, 2n - 2k$

Remark 5.1 (the dichotomy: a heuristic, not a theorem). A pure harmonic monomial $\prod_t H_{x_t}^{(r_t)}$ produces, in the limit, values in the $\mathbb{Q}$ -algebra generated by $\zeta(2), \zeta(3), \dots$ and $\log 2$ (the last from letters at wide arguments, e.g. $H_{2k}^{(1)} - H_k^{(1)} \rightarrow \log 2$). If $L(\chi, w)$ lies outside that algebra — which is expected but open, already for Catalan’s constant $G = L(\chi_{-4}, 2)$, whose irrationality is not known — then no χ -twisted family can have a pure- ζ harmonic decomposition, and its letters must be twisted. We label this as what it is: an expectation resting on unavailable transcendence input. It is not used in the proof of any theorem below; it is the explanation the data invites.

This is confirmed by the decisive experiment on **E**: [VERIFIED: bases of up to 44 elements over 8 argument forms] **E**’s fitting system is *inconsistent in every pure harmonic basis tried*, and becomes consistent the moment χ_{-4} -letters are admitted, yielding the five-term weight above. **E** is the only known instance of the twisted case $e = 1$ of Theorem 4.5.

5.2. The four theorems.

Theorem 5.2. [THM] *Congruence (1.2) holds for γ and **A** with $\chi \equiv 1$ and all $p \geq 5$, and for **D** and s_{10} with $\chi \equiv 1$ and all $p \geq 7$.*

Proof. Apply Theorem 4.5 with the weights of §5.1. (H5) holds with $e = 0$. (H4): the arguments are among $n, k, n - k$, all in $[0, n]$; the coefficients have denominators 3, 6 (γ), 4 (**A**), 5 (**D**), 5 (s_{10}), whence the stated prime bounds. (H2)’s Δ -clause is free by Lemma 4.4. For (H1)–(H3), write

$S = \binom{n}{k}^\alpha \binom{n+k}{n}^\beta$: the surviving set is $\{s \leq r\}$ (no borrow in $\binom{n}{k}$) intersected with $\{r + s < p\}$ (no carry in $\binom{n+k}{n}$); this second constraint is vacuous for $\mathbf{A}$ and s_{10} , where $\beta = 0$), a product region; there $\binom{n}{k} \equiv \binom{a}{b} \binom{r}{s}$ and $\binom{n+k}{n} \equiv \binom{a+b}{a} \binom{r+s}{r}$ by Lucas, and the deleted terms with $r + s \geq p$ vanish mod p , so $\sum_{s \in \Sigma_r} S(r, s) \equiv A(r)$. Digit compatibility is $\lfloor k/p \rfloor = b$, $\lfloor (n-k)/p \rfloor = a - b$ (no borrow), $\lfloor n/p \rfloor = a$. The details are [21, §3]. $\square$

[VERIFIED: 0 failures, all cells $n = ap + r < p^2$] both layers of the proof separately: γ , $\mathbf{A}$, s_{10} at $p = 5, 7, 11, 13$; $\mathbf{D}$ at $p = 7, 11, 13$, with 11 failing cells at $p = 5$ — exactly the (H4) coefficient condition, $\mathbf{D}$'s coefficients being $\frac{1}{5}(1, 2, -1, -1)$. So $\mathbf{D}$'s proof is unconditional for $p \geq 7$ and $p = 5$ is [VERIFIED] only. (s_{10} 's coefficients are likewise $\frac{1}{5}(1, 4, -4)$; it happens to show no $p = 5$ defect, but the proof as written still requires $p \geq 7$.) *No other defect of any kind appears in the tame cases: the only obstruction the mechanism ever meets is the coefficient condition.*

5.3. s_7 : **reachable**. For s_7 , ε , α and $\mathbf{E}$ the known weight is *not* tame: arguments reach $2n$ (letters at $2k$, $2n - 2k$, $2k - n$), so for k in the vanishing layer $\lfloor x/p \rfloor$ can reach $2p - 1$, the descended letter acquires a pole, and $p^w \mathfrak{w}(n, k)$ is no longer p -integral. The two-layer split of the proof of Theorem 4.5 then fails *layer by layer*: each of

$$V(n) = \sum_{k: p \mid S(n, k)} S(n, k) p^w \mathfrak{w}(n, k), \quad U(n) = \sum_{k: p \nmid S(n, k)} S(n, k) p^w \mathfrak{w}(n, k)$$

separately misses its intended value. (It does *not* fail index by index in k ; that misreading led to a false claim in an earlier memo of the programme, corrected in [24].)

Proposition 5.3. [VERIFIED] *For s_7 and every prime $p \in \{5, 11, 13, 17, 19, 23, 29, 31, 37, 41\}$, all cells $n = ap + r < p^2$ and all summation indices k :*

$$(V1) \quad v_p(S(n, k)) + v_p(p^2 \mathfrak{w}(n, k)) \geq 1 \quad \text{whenever } p \mid S(n, k),$$

with minimum exactly 1 (1 659 068 vanishing-layer terms, 0 violations); and

$$(V2) \quad v_p(S(a, b)) + v_p(\mathfrak{w}(a, b)) \geq 1 \quad \text{whenever } p \mid S(a, b), \quad a < p,$$

0 violations for $p \in \{5, 11, 13, 17, 19, 23, 29, 31, 37\}$.

(V1) implies that the vanishing layer is $\equiv 0 \pmod{p}$ and (V2) that $\Delta(a) \equiv 0 \pmod{p}$, so at each prime at which (V1), (V2), (H1), (H3) and the product structure are known — all [VERIFIED: $5 \leq p \leq 23$, 0 failures] — the proof of Theorem 4.5 goes through for s_7 . At those primes (1.2) was in any case verified directly, so what *is* gained is not a fact but a route: a proof of (V1) and (V2) for all $p \geq 5$, $p \neq 7$ would make s_7 a theorem. We state this as a problem and not as a proposition.

The prime $p = 7$ is excluded for s_7 : its coefficients have denominator 28, and there are 13 failing cells at $p = 7$ against none at $p = 5, 11, 13$. The reason (V1) is true for s_7 and false for the other three is structural: in s_7 's weight the argument $2k$ occurs *only inside weight-one letters*, so the pole has order at most 1 and a single Kummer carry in $\binom{2k}{n}$ beats it. This is a bounded, purely local Kummer-versus-order-one-pole problem.

Problem 5.4. [OPEN] Prove (V1) and (V2) for s_7 , $p \geq 5$, $p \neq 7$. This would take the proved count from four to five. It does *not* deliver an instance with $\chi \neq 1$: s_7 has $\chi = 1$.

5.4. α , ε , $\mathbf{E}$: **blocked, and provably so.**

Proposition 5.5 (the defect is real). [VERIFIED] *For α , ε and $\mathbf{E}$, the inequality $v_p(S(n, k)) + v_p(p^w \mathfrak{w}(n, k)) \geq 1$ on $\{p \mid S(n, k)\}$ is false, with explicit counterexamples, and so is its aggregated weakening $v_p(V(n)) \geq 1$.*

The minimal certificates, in exact rational arithmetic, are:

- α (Domb), $p = 5$, $n = 20 = 4 \cdot 5 + 0$, $k = 5$: $S = \binom{20}{5}^2 \binom{10}{5} \binom{30}{15}$ has $v_5(S) = 1$, while $v_5(\mathfrak{w}(20, 5)) = -5$, so $v_5(p^3 \mathfrak{w}) = -2$ and the sum is $-1 < 1$. The pole sits at $2n - 2k = 30 \geq p^2$, through the monomials $H_k^{(1)} (H_{2n-2k}^{(1)})^2$ and $H_k^{(2)} H_{2n-2k}^{(1)}$; the term $S p^w \mathfrak{w}$ is not even p -integral.
- ε , $p = 5$, $n = 15$, $k = 15$: $v_5(S) = 2$, $v_5(p^3 \mathfrak{w}) = -2$, sum $0 < 1$; pole at $2k = 30$, through $H_{2k}^{(3)}$ and $H_{2k}^{(2)} H_{2k}^{(1)}$.
- $\mathbf{E}$, $p = 5$, $n = 15$, $k = 15$: $v_5(S) = 1$, $v_5(p^2 \mathfrak{w}) = -1$, sum $0 < 1$; pole at $2k = 30$, through $K_{\chi-4}^{(2)}(2k)$ and $H_{2k}^{(1)} K_{\chi-4}^{(1)}(2k)$.

The observed minima are uniform in n and p (-1 for α , 0 for ε and $\mathbf{E}$, [VERIFIED: $5 \leq p \leq 23$]), so a *corrected* inequality with a constant amount of slack does exist — and is useless, because the proof needs ≥ 1 exactly. Note the sharpening that ε 's summand $\binom{n}{k}^2 \binom{2k}{n}^2$ is a square, so $v_p(S) \in 2\mathbb{Z}$ and one Kummer carry already buys two orders; that is why ε lands at 0 rather than -2 .

Proposition 5.6 (what is true instead). [VERIFIED] *For all four non-tame families, all cells $n = ap + r < p^2$ and $p \in \{5, 7, 11, 13, 17, 19, 23\}$ ($\alpha, \varepsilon, \mathbf{E}$) resp. $p \in \{5, 11, \dots, 37\}$ (s_7), 0 failures in 20/20, 42/42, 110/110, 156/156, 272/272, 342/342, 506/506 cells for both lines:*

$$V(n) \equiv \chi(p)^e \Delta(a) A(r) \pmod{p}, \quad (\text{LD})$$

$$U(n) \equiv \chi(p)^e (B(a) - \Delta(a)) A(r) \pmod{p}. \quad (\text{LD}')$$

Adding the two lines returns (1.2). So (LD) is exactly the missing ingredient, and nothing else is missing.

This is the precise content of the earlier, vaguer statement that “the vanishing-layer and surviving-layer defects occur in equal numbers and cancel”. They are equal in number because they are the *same* number, $\chi(p)^e \Delta(a) A(r)$, with opposite signs. We stress that “equal numbers” was never independent evidence: given the (verified) conclusion (1.2), V fails if and only if U does. The genuine new information is the closed form. Certificates: α at $p = 5$, $n = 20$: $V \equiv \Delta(4) A(0) = 37259/54$, matched at $v_5 = 3$; $\mathbf{E}$ at $p = 7$, $n = 28$: $\Delta(4) = 379/9$, $\chi_{-4}(7) = -1$, $V \equiv -379/9$, matched at $v_7 = 2$. Note also that $\Delta(a) = 0$ identically for $a \leq (p-1)/2$ — the defect is a large-digit phenomenon.

Theorem 5.7 (no admissible weight repairs the defect). [EXCLUDED] *Let $\mathcal{K}$ be the exact $\mathbb{Q}$ -kernel of the fitting system for one of $\alpha, \varepsilon, \mathbf{E}$ over the argument alphabets below, so that the admissible weights form the affine space $\mathfrak{w}_0 + \mathcal{K}$. Then, for each of the primes $p \in \{5, 7, 11, 13, 17, 19, 23\}$ marked $\times$ below, the $\mathbb{F}_p$ -linear system $\sum_{b \in \mathcal{B}_a} S(a, b) \mathfrak{w}(a, b) \equiv B(a)$ ($a = 1, \dots, p-1$), i.e. $\Delta(a) \equiv 0$ for all a , is inconsistent:*

family	alphabet	cols	dim $\mathcal{K}$	5	7	11	13	17	19	23
α	$\{k, n-k, 2k, 2n-2k\}$	40	21	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$
α	$+n$	65	32	$\checkmark$	$\checkmark$	$\times$	$\times$	$\times$	$\times$	$\times$
ε	$\{k, n-k, 2k, 2k-n\}$	40	2	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$
ε	$+n, 2n-2k$	98	7	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$
$\mathbf{E}$	$\{k, n-k, 2k, 2n-2k\} + \chi_{-4}, (H5)$	20	10	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$
$\mathbf{E}$	$+n, (H5) \text{ dropped}$	65	28	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$
s_7	$\{n, k, n-k, 2k\} (+\text{enlarged})$	14/27	2	$\checkmark$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

For $\mathbf{E}$ the kernel has rank 0 against Δ at each of those primes: every admissible weight produces the same $\Delta(a) \bmod p$.

Remark 5.8. The obvious extrapolation — that $\Delta(a) \bmod p$ is an invariant of $\mathbf{E}$ at every prime, not an artefact of the fit — is not proved here and is recorded as such. Nothing below depends on

it: what the theorem already excludes is the strategy of re-choosing the weight, and it excludes it at every prime that has been examined.

Method and scope. The kernel basis is computed exactly (row reduction modulo four primes, CRT, rational reconstruction) and each basis vector is verified over $\mathbb{Q}$ by $\sum_k S(n, k)\kappa(n, k) = 0$ for $n = 1, \dots, 21$ (0 failures in every case); the $\mathbb{F}_p$ system is then solved exactly. The **E** statement is confirmed independently by direct perturbation: at $p = 5$ all ten kernel vectors leave $(\Delta(3), \Delta(4)) \equiv (3, 1) \pmod{5}$ unchanged; for α only two kernel vectors move it, and only along $(1, -1)$, never onto $(0, 0)$. *Scope, stated honestly:* this excludes every weight-homogeneous $\mathbb{Q}$ -combination of bare letters at the listed argument forms. It does not exclude a wholly different ansatz — rational-function coefficients, a constant-term representation, or new letters. $\square$

Corollary 5.9. *The route “prove one valuation inequality and move $\alpha, \varepsilon, s_7, \mathbf{E}$ inside Theorem 4.5” — proposed in an earlier memo of the programme and repeated in [21, §9] — is wrong. The inequality is false for $\alpha, \varepsilon, \mathbf{E}$; it works for s_7 alone, taking the proved count from four to five, not from four to eight. For $\alpha, \varepsilon, \mathbf{E}$ the target is (LD), which is a descent congruence of the same logical type as the theorem it is meant to prove, not a local Kummer estimate. That is a materially larger gap, and it is where the programme’s first instance with $\chi \neq 1$ is gated.*

Two structural leads survive. First, $\Delta(a) = 0$ identically for $a \leq (p-1)/2$. Second, the per-digit refinement $V_b(n) \equiv \chi(p)^e S(a, b)\mathfrak{w}(a, b)A(r)$ of (LD) is [VERIFIED: 0 failures, $p = 5, 7, 11, 13$] for $\varepsilon, \mathbf{E}$ and s_7 , so for those (LD) is provable one digit at a time; it is false for α (e.g. 16 of 30 defect-digits fail at $p = 5$), so α needs the aggregated form.

Finally, **E** is provably non-tame in a stronger sense: attempts to find a tame χ_{-4} -decomposition for **E** over the complete tame argument set $\{n, k, n-k\}$, with χ_{-4} , with three characters, with and without (H5), are all [EXCLUDED] at 106–154 excess equations; the only other tame representation of **E**’s first row that exists, $A(n) = \sum_k 4^{n-2k} \binom{n}{2k} \binom{2k}{k}^2$ ([VERIFIED: $n \leq 30$]), supports no decomposition at all (230 columns, rank 230, 90 excess); and a sweep of 4.4 million tame templates finds no third representation. The wide arguments are essential, so the twisted case of Theorem 4.5 is proved as a mechanism but has, at present, no tame instance.

5.5. The conjectural seven. For **B, C, F, $\delta, \zeta, \eta, s_{18}$** no harmonic decomposition is known and (LB_w^χ) is [VERIFIED] only. The searches were run to completion in every *complete tame* alphabet (the set of tame integer linear forms on a given support is finite and explicitly computable), at up to 520 columns and with 55 to 193 excess equations; all 30 runs are inconsistent, and each headline negative was recomputed at a second, unrelated modulus with identical rank and verdict. Two of these are worth stating as results.

Proposition 5.10 (C). [EXCLUDED] **C**’s second solution admits no harmonic-monomial decomposition on the summand $\binom{n}{k}^2 \binom{2k}{k}$ in any basis tried: plain harmonic monomials at $\{n, k, n-k, 2k\}$ (14 elements) and at $\{n, k, n-k, 2k, n+k, 2n-2k, 2n, 2n-k\}$ (44), and with χ_{-3} -letters added (44 and 152 elements, 110 and 220 exact equations); and, sharper, not even when the conductor-three arguments $3k, 3n-3k, 3n$ that the summand does not supply are furnished by hand (90 columns / 64 excess and 132 / 63). By the heuristic of Remark 5.1 its letters are expected to be twisted; the obstruction is therefore not merely the missing arguments. Moreover **C**’s complete tame form set is only $\{n, k, n-k\}$ (its $2k$ reaches $2n$), so Theorem 4.5 could not be applied to **C** even if a decomposition were found in a wide alphabet.

Proposition 5.11 (δ). [EXCLUDED] δ ’s second solution admits no weight-three harmonic decomposition over its complete tame alphabet $\{n, k, 2k, 3k, n-k, n-2k, n-3k\}$ at full degree (140 columns, 60 excess; also 330 columns / 70 excess with χ_{-3} -letters), and δ possesses no tame binomial representation whatsoever in a 4.4-million-template sweep.

A useful diagnostic emerged from these runs: the rank deficiency of the design matrix. Families that do have a decomposition have large kernels (**E**: rank 37 of 65; α : 19 of 40); **C**, **F**, s_{18} , η have small ones (86/90, 100/152, 107/110, 153/156); and **B**, δ , ζ have *rank = columns in every run* — no $\mathbb{Q}$ -relation at all between their summand and the harmonic letters. So enlarging the alphabet will not help those three; the ansatz has to change. The recommended replacement is Gorodetsky's constant-term representation [10], which supplies exactly the sort of new letters that Proposition 5.10 shows are needed.

The open coverage gaps, stated rather than hidden: degree-three-in-weight-one coverage for ζ and η in their complete tame spaces, and for **F** at seven tame arguments, would need 424–1088 columns and correspondingly 500–1150 rows; those three runs at full degree were not made.

6. CLOSED FORMS

6.1. Apéry's $\zeta(3)$ numerators.

Theorem 6.1. [THM] [LEAN] *Let b_n be defined by $b_0 = 0$, $b_1 = 6$ and $(n+2)^3 b_{n+2} = (34n^3 + 153n^2 + 231n + 117)b_{n+1} - (n+1)^3 b_n$, i.e. Apéry's recurrence. Then for every $n \geq 0$*

$$b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 (2H_n^{(3)} - H_k^{(3)}).$$

The proof is in [22]; it is elementary — two Zeilberger/Gosper certificates that are polynomial identities in $\mathbb{Q}[n, k]$, combined into a *single* telescope with vanishing boundary values — and it is formalised (§8). It replaces both Apéry's own weight $H_n^{(3)} + \sum_{m \leq k} (-1)^{m-1} (2m^3 \binom{n}{m} \binom{n+m}{m})^{-1}$ and Nesterenko's weight $H_k^{(3)} + (2H_k^{(1)} - H_{n+k}^{(1)} - H_{n-k}^{(1)})H_k^{(2)}$ [12]: the fitting system has an 11-dimensional kernel, so the harmonic-monomial representation is far from unique, and this is the shortest member. We have not located it in the literature; see [22, §9] for the search record.

6.2. The symmetric template. For the d -fold sums $A^{(d)}(n) = \sum_k \binom{n}{k}^d$ the summand is symmetric under $k \mapsto n-k$, so any fitted weight may be replaced by its $k \leftrightarrow n-k$ average without changing $B(n)$. Writing

$$\sigma(n, k) = H_k^{(2)} + H_{n-k}^{(2)}, \quad \delta(n, k) = H_k^{(1)} - H_{n-k}^{(1)}, \quad (6.1)$$

the fitted weights of §5.1 symmetrise to $\mathfrak{w} = \alpha_d \sigma + \beta_d \delta^2$ with $\alpha_d = \frac{1}{2(d+1)}$, $\beta_d = \frac{d}{2(d+1)}$; note $\alpha_d + \beta_d = \frac{1}{2}$ for every d .

This is not cosmetic. In the symmetric form the only weight-one letter that ever appears is the single antisymmetric combination δ , because σ differences to *rational* functions under $n \mapsto n \pm 1$ and $k \mapsto k + 1$:

$$\begin{aligned} \sigma(n+1, k) - \sigma(n, k) &= \frac{1}{(n+1-k)^2}, & \delta(n+1, k) - \delta(n, k) &= \frac{-1}{n+1-k}, \\ \sigma(n-1, k) - \sigma(n, k) &= \frac{-1}{(n-k)^2}, & \delta(n-1, k) - \delta(n, k) &= \frac{1}{n-k}, \\ \sigma(n, k+1) - \sigma(n, k) &= \frac{1}{(k+1)^2} - \frac{1}{(n-k)^2}, & \delta(n, k+1) - \delta(n, k) &= e(n, k) := \frac{1}{k+1} + \frac{1}{n-k}. \end{aligned} \quad (6.2)$$

So the certificate module is free of rank 2 on $\{1, \delta\}$ instead of rank 4 on $\{1, H_k^{(1)}, H_{n-k}^{(1)}, H_n^{(1)}\}$: *two* Gosper problems instead of four.

The template. Let $c_+(n)u_{n+1} + c_0(n)u_n + c_-(n)u_{n-1} = 0$ be the recurrence and $S(n, k) = \binom{n}{k}^d$.

- (1) *Zeilberger certificate.* Seek $\rho \in \mathbb{Q}(n)[k]$ with $G(n, k) = S(n, k)k^d \rho(n, k)/(n+1-k)^d$ satisfying

$$c_+ S(n+1, k) + c_0 S(n, k) + c_- S(n-1, k) = G(n, k+1) - G(n, k). \quad (6.3)$$

Because $S(n, k+1)/S(n, k) = ((n-k)/(k+1))^d$, one has $G(n, k+1) = S(n, k)\rho(n, k+1)$: *the shifted certificate is the summand times a polynomial.* This is what makes the scheme work, and it turns (6.3) into one identity in $\mathbb{Q}(n, k)$.

- (2) *Split and Abel.* With $\Delta^\pm$ the n -differences (6.2),

$$L[B](n) = \sum_{k=0}^n \mathfrak{w}(n, k)(G(n, k+1) - G(n, k)) + \sum_k \text{Corr}(n, k) + \text{Top}(n),$$

$\text{Corr} = c_+ S(n+1, k)\Delta^+ \mathfrak{w} + c_- S(n-1, k)\Delta^- \mathfrak{w}$ and $\text{Top}(n) = c_+(n)S(n+1, n+1)\mathfrak{w}(n+1, n+1)$; Abel summation and $G(n, 0) = 0$ turn the first sum into $\mathfrak{w}(n, n)G(n, n+1) - \sum_{k=0}^{n-1} G(n, k+1)\Delta_k \mathfrak{w}$.

- (3) *Two Gosper problems.* By (6.2) the residual k -summand is $S(n, k)(\mathcal{A}(n, k) + \mathcal{B}(n, k)\delta(n, k))$ with $\mathcal{A}, \mathcal{B} \in \mathbb{Q}(n, k)$. Seek $Z(n, k) = S(n, k)(z_0 + z_1\delta)$ with $Z(n, k+1) - Z(n, k)$ equal to it. Since $\delta(n, k+1) = \delta(n, k) + e(n, k)$, matching coefficients of 1 and δ gives, with $r = S(n, k+1)/S(n, k)$,

$$r(n, k) z_1(n, k+1) - z_1(n, k) = \mathcal{B}(n, k), \quad (6.4)$$

$$r(n, k) z_0(n, k+1) - z_0(n, k) = \mathcal{A}(n, k) - r(n, k)z_1(n, k+1)e(n, k). \quad (6.5)$$

The ansatz $z_1 = N(k)/(n+1-k)^{d+1}$, $z_0 = M(k)/(n+1-k)^{d+2}$ reduces both to linear systems over $\mathbb{Q}(n)$.

- (4) *Boundary.* Everything cancels iff $Z(n, 0) = 0$ and three scalar identities hold.

Proposition 6.2 (the boundary identities, for every d). [PROVED] *With $\mathfrak{w} = \alpha\sigma + \beta\delta^2$, and provided (6.3), (6.4), (6.5) and $Z(n, 0) = 0$ hold, one has $L[B](n) = 0$ for all $n \geq 1$ as soon as*

$$\begin{aligned} \rho(n, n+1) &= -c_+(n), & N(n) &= 2\beta c_+(n) \left((n+1)^d - \frac{1}{n+1} \right), \\ M(n) &= -c_+(n)(\alpha + \beta) \left((n+1)^d + \frac{1}{(n+1)^2} \right). \end{aligned} \quad (6.6)$$

In particular, for $c_+(n) = (n+1)^{d-1}$, $\alpha = \alpha_d$, $\beta = \beta_d$,

$$\rho(n, n+1) = -(n+1)^{d-1}, \quad N(n) = \frac{d}{d+1} \left((n+1)^{2d-1} - (n+1)^{d-2} \right),$$

$$M(n) = -\frac{1}{2} \left((n+1)^{2d-1} + (n+1)^{d-3} \right).$$

Proof. Assemble step (2)–(4). Since $S(n-1, k) = 0$ for $k \geq n$ and $S(n, n) = S(n+1, n+1) = 1$,

$$L[B](n) = Z(n, n) - Z(n, 0) + \text{Corr}(n, n) + \mathfrak{w}(n, n)G(n, n+1) + \text{Top}(n).$$

Now $\sigma(n, n) = H_n^{(2)}$, $\delta(n, n) = H_n^{(1)}$, $G(n, n+1) = \rho(n, n+1)$, $Z(n, n) = M(n) + N(n)H_n^{(1)}$, $\text{Corr}(n, n) = c_+(n+1)^d[\alpha + \beta(1 - 2H_n^{(1)})]$ and $\text{Top}(n) = c_+[\alpha H_{n+1}^{(2)} + \beta(H_{n+1}^{(1)})^2]$. Substituting $H_{n+1}^{(2)} = H_n^{(2)} + (n+1)^{-2}$ and $H_{n+1}^{(1)} = H_n^{(1)} + (n+1)^{-1}$ and collecting the coefficients of $H_n^{(2)}$, $(H_n^{(1)})^2$, $H_n^{(1)}$ and 1 shows that (6.6) makes each of the four coefficients vanish identically — the first two both reduce to $\rho(n, n+1) + c_+(n) = 0$ — whence $L[B](n) = 0$. The last statement follows by substituting $\alpha_d + \beta_d = \frac{1}{2}$. $\square$

Remark 6.3. The conditions (6.6) are also necessary, since $H_n^{(2)}$, $(H_n^{(1)})^2$, $H_n^{(1)}$ and 1 are linearly independent over $\mathbb{Q}(n)$ — a standard fact of $\Pi\Sigma$ -difference field theory, cf. [14] — but we do not use necessity and do not prove it here.

Remark 6.4. Proposition 6.2 says the boundary is *never* the obstruction: the three identities hold for every d . The only d -dependent step is the existence of a rational solution of (6.4)–(6.5). It is worth noting where this leaves the general d : Chamberland–Straub’s conjecture [6, Conj. 9] was proved for every d by Straub–Zudilin [17] by an entirely different route (Remark 6.8), so the template is not needed for it; what the template supplies, where it closes, is a *finite certificate* valid for every $n \geq 1$.

6.3. Franel and $\sum_k \binom{n}{k}^4$: finite certificates. The two closed forms below are not new — see Remark 6.8 — but the proofs are, and it is the harmonic-monomial *display* that §5 needs.

Theorem 6.5 (Franel). [THM] *Let $A(n) = \sum_k \binom{n}{k}^3$ ($A000172$), so $(n+1)^2 u_{n+1} = (7n^2 + 7n + 2)u_n + 8n^2 u_{n-1}$. Then the second solution is*

$$B(n) = \sum_{k=0}^n \binom{n}{k}^3 \left[\frac{1}{8} (H_k^{(2)} + H_{n-k}^{(2)}) + \frac{3}{8} (H_k^{(1)} - H_{n-k}^{(1)})^2 \right].$$

Theorem 6.6 (s_{10}). [THM] *Let $A(n) = \sum_k \binom{n}{k}^4$, so $(n+1)^3 u_{n+1} = (2n+1)(6n^2 + 6n + 2)u_n + n(64n^2 - 4)u_{n-1}$. Then the second solution is*

$$B(n) = \sum_{k=0}^n \binom{n}{k}^4 \left[\frac{1}{10} (H_k^{(2)} + H_{n-k}^{(2)}) + \frac{2}{5} (H_k^{(1)} - H_{n-k}^{(1)})^2 \right].$$

Proof of Theorems 6.5 and 6.6. Both sides vanish at $n = 0$ and equal 1 at $n = 1$, so it suffices to prove that the right-hand side is annihilated by the recurrence operator for $n \geq 1$; the leading coefficient $(n+1)^{d-1}$ never vanishes, so the second solution is then determined. Apply the template of §6.2 with the certificates

$$\begin{aligned} \rho_F(n, k) &= -\frac{1}{n} (4 - 12k + 12k^2 - 4k^3 + 22n - 39kn + 18k^2n + 32n^2 - 27kn^2 + 14n^3), \\ N_F(k) &= \frac{3k^2}{4n} (4 - 16k + 24k^2 - 16k^3 + 4k^4 + 26n - 68kn + 63k^2n - 20k^3n \\ &\quad + 54n^2 - 88kn^2 + 39k^2n^2 + 46n^3 - 36kn^3 + 14n^4), \\ M_F(k) &= -\frac{k}{2n} (2 - 10k + 20k^2 - 20k^3 + 10k^4 - 2k^5 + 15n - 50kn + 70k^2n - 45k^3n \\ &\quad + 11k^4n + 40n^2 - 90kn^2 + 80k^2n^2 - 25k^3n^2 + 50n^3 - 70kn^3 + 30k^2n^3 \\ &\quad + 30n^4 - 20kn^4 + 7n^5) \end{aligned}$$

for $d = 3$, and the analogous ρ_{10} (degree 5 in k , degree 6 in n), N_{10} , M_{10} (degree 9 in k) for $d = 4$, recorded in the data file accompanying [26]. Every one of (6.3), (6.4), (6.5) is an identity in $\mathbb{Q}(n, k)$ and was verified symbolically, by clearing denominators and expanding to 0; $z_1(n, 0) = z_0(n, 0) = 0$ because N and M carry factors k^2 , k ($d = 3$) resp. k^3 , k^2 ($d = 4$). The three boundary identities (6.6) hold: $\rho_F(n, n+1) = -(n+1)^2$, $N_F(n) = \frac{3}{4}((n+1)^5 - (n+1))$, $M_F(n) = -\frac{1}{2}((n+1)^5 + 1)$; and $\rho_{10}(n, n+1) = -(n+1)^3$, $N_{10}(n) = \frac{4}{5}((n+1)^7 - (n+1)^2)$, $M_{10}(n) = -\frac{1}{2}((n+1)^7 + (n+1))$ — exactly the values forced by Proposition 6.2. Proposition 6.2 then gives $L[B] = 0$. $\square$

Remark 6.7 (certificate audit). The identities (6.3), (6.4), (6.5), $Z(n, \cdot)$ at $k = 0$, and (6.6) were each re-derived and re-verified symbolically for the present paper, independently of the memo they come from, in a computer-algebra system, each returning 0 after simplification. In addition $L[B] = 0$ was checked in exact rational arithmetic for $n \leq 15$ for both families, and $B(0) = 0$, $B(1) = 1$ by hand.

Remark 6.8 (these sums are Straub–Zudilin’s t^2 -coefficients). Straub–Zudilin [17] study the analytic deformation

$$A^{(s)}(n, t) = \sum_{k=0}^n \binom{n}{k}^s \prod_{j=1}^k \left(1 - \frac{t}{j}\right)^{-s} \prod_{j=1}^{n-k} \left(1 + \frac{t}{j}\right)^{-s} = \sum_{j \geq 0} A_j^{(s)}(n) t^{2j},$$

and prove (their Theorem 1.3) that every $A_j^{(s)}(n)$ solves the telescoping recurrence of $A^{(s)}(n)$ for large enough n , with $A_1^{(s)}(1) = s(s+1)$. Since

$$\prod_{j \leq k} \left(1 - \frac{t}{j}\right)^{-s} \prod_{j \leq n-k} \left(1 + \frac{t}{j}\right)^{-s} = \exp\left(st\delta + \frac{s}{2}t^2\sigma + O(t^3)\right) = 1 + st\delta + t^2\left(\frac{s}{2}\sigma + \frac{s^2}{2}\delta^2\right) + O(t^3)$$

with σ, δ as in (6.1), one gets $A_1^{(s)}(n)/(s(s+1)) = \sum_k \binom{n}{k}^s (\alpha_s \sigma + \beta_s \delta^2)$ identically. *So Theorems 6.5 and 6.6 are the cases $s = 3, 4$ of [17, Thm. 1.3], written out in harmonic numbers ([VERIFIED: exact, $s = 3, 4, 5, 6$, $n \leq 12$]).* What the proof above adds is that the identity holds for *every* $n \geq 1$ — [17] obtains the recurrence only for n large — by a finite certificate rather than a locally uniform analytic limit; that certificate is what makes the statement formalisable in the sense of §8.

Corollary 6.9 (Apéry limits, after Straub–Zudilin). [PROVED] *For $d = 3$ and $d = 4$,*

$$\lim_{n \rightarrow \infty} \frac{B^{(d)}(n)}{A^{(d)}(n)} = \frac{\zeta(2)}{d+1},$$

i.e. $\pi^2/24$ and $\pi^2/30$.

Proof. Write $m = \lfloor n/2 \rfloor$ and $T = \lceil n^{3/4} \rceil$. By Hoeffding’s bound $\binom{n}{k} \leq 2^n e^{-2(k-n/2)^2/n}$ and $\binom{n}{m} \geq 2^n/(n+1)$, so $\binom{n}{k} \leq (n+1)\binom{n}{m} e^{-2(k-n/2)^2/n}$ for all k . Hence, with $A^{(d)}(n) \geq \binom{n}{m}^d$,

$$\frac{1}{A^{(d)}(n)} \sum_{|k-n/2| \geq T} \binom{n}{k}^d \leq (n+1)^d \sum_{|u| \geq T} e^{-2du^2/n} \leq \frac{(n+1)^d n}{d(T-1)} e^{-2d(T-1)^2/n} = O(e^{-\sqrt{n}}),$$

the sum over $u \in \frac{1}{2}\mathbb{Z}$ being bounded by twice the corresponding integral. On the complementary range $|k-n/2| < T$ we have, for n large, $k \geq n/3$ and $n-k \geq n/3$, whence $|H_k^{(2)} - \zeta(2)| \leq 1/k \leq 3/n$ and likewise for $n-k$, so $|\sigma(n, k) - 2\zeta(2)| \leq 6/n$; and, assuming $k \geq n-k$ without loss of generality, $|\delta(n, k)| = \sum_{j=n-k+1}^k j^{-1} \leq (2k-n)/(n-k) \leq 6n^{-1/4}$, so $\delta(n, k)^2 \leq 36n^{-1/2}$. Globally $0 \leq \sigma \leq 2\zeta(2)$ and $|\delta| \leq H_n^{(1)} \leq 1 + \log n$. Therefore

$$\left| \frac{B^{(d)}(n)}{A^{(d)}(n)} - 2\alpha_d \zeta(2) \right| \leq \frac{6\alpha_d}{n} + 36\beta_d n^{-1/2} + O(e^{-\sqrt{n}}) (4\alpha_d \zeta(2) + \beta_d (1 + \log n)^2) \rightarrow 0,$$

and $2\alpha_d \zeta(2) = \zeta(2)/(d+1)$. Theorems 6.5 and 6.6 identify the sums with the second solutions normalised as in Definition 2.1, which is the normalisation of [6, eq. (20)]. $\square$

Remark 6.10. Both limits were observed numerically by Cusick and are reported by van der Poorten [18, p. 202]; the case $d = 3$ was proved by Zagier [20] using modular forms, and *both cases, indeed all d* , are Example 1.5 of Straub–Zudilin [17], which settles [6, Conj. 9]. Corollary 6.9 is therefore a re-proof, not a new result. It is nevertheless worth recording, for two reasons: it is elementary, using no modular forms and no analytic deformation, only the closed form plus a Chernoff estimate; and it exhibits the mechanism — the summand $\binom{n}{k}^d$ concentrates at $k = n/2$, where $\delta(n, k) \rightarrow 0$ and $\sigma(n, k) \rightarrow 2\zeta(2)$, so that the Apéry limit is simply $2\alpha_d \zeta(2)$ and the δ^2 -part contributes nothing. The argument does not reach $d \geq 5$: the concentration estimate is uniform in d , but the minimal recurrence then has order > 2 and the closed form of §6.2 is not available. *Equation and conjecture numbers quoted from [6] are those of its arXiv version; the journal version may renumber.*

6.4. A provable insufficiency: the $\zeta(2)$ -Apéry pair.

Observation 6.11. [VERIFIED: $n \leq 25$] For $\mathbf{D}$, $A(n) = \sum_k \binom{n}{k}^2 \binom{n+k}{k}$ (A005258), $(n+1)^2 u_{n+1} = (11n^2 + 11n + 3)u_n + n^2 u_{n-1}$, the second solution is

$$B(n) = \frac{1}{5} \sum_k \binom{n}{k}^2 \binom{n+k}{k} \left[H_n^{(2)} + H_k^{(1)} (2H_k^{(1)} - H_{n-k}^{(1)} - H_n^{(1)}) \right],$$

with $B(0), \dots, B(6) = 0, 1, \frac{25}{4}, \frac{1741}{36}, \frac{6585}{16}, \frac{13327519}{3600}, \frac{124308457}{3600}$.

Steps (1)–(3) of the template go through, and beautifully. The Zeilberger certificate is the smallest in the programme,

$$\rho_D(n, k) = k^2 + k(1 + 6n) - 4 - 15n - 11n^2, \quad G_D(n, k) = \frac{k^3 \rho_D(n, k) \binom{n+1}{k}^2 \binom{n+k-1}{k}}{n(n+1)^2},$$

[PROVED] (a rational identity, expanded to 0; re-verified here). The structural boundary identity holds *exactly*: $\rho_D(n, n+1) = -2(n+1)(2n+1) = -(n+1)^2 \binom{2n+2}{n+1} / \binom{2n}{n} = -c_+(n)S(n+1, n+1)/S(n, n)$. The correction decomposes as $Y(n, k)/S(n, k) = \mathcal{A} + \mathcal{B}_1 H_k^{(1)} + \mathcal{B}_2 (H_{n-k}^{(1)} + H_n^{(1)})$ with $\mathcal{B}_2 = \rho_D(n, k+1)/(5(k+1))$ [VERIFIED: exact, $n = 2, \dots, 8$].

Theorem 6.12 (the $H_{n-k}^{(1)}$ component is not Gosper-summable). [PROVED] *Regard n as an indeterminate and put $T(k) = S_D(n, k)\mathcal{B}_2(n, k)$. Then $T(k+1)/T(k) = (a(k)/b(k))(c(k+1)/c(k))$ with $a = (n-k)^2(n+k+1)$, $b = (k+1)^2(k+2)$, $c(k) = \rho_D(n, k+1)$, and $\gcd(a(k), b(k+j)) = 1$ in $\mathbb{Q}(n)[k]$ for every $j \geq 0$ (the roots of a are n and $-n-1$, those of $b(k+j)$ are $-j-1$ and $-j-2$), so this is Gosper's normal form. Gosper's key equation $a(k)x(k+1) - b(k-1)x(k) = c(k)$ has no solution $x \in \mathbb{Q}(n)[k]$. Consequently T has no hypergeometric antidifference with certificate rational in n , which is what a proof uniform in n requires.*

Proof. Write $A = a(k)$, $B = b(k-1)$ and use

$$Ax(k+1) - Bx(k) = \frac{1}{2}(A-B)(x(k+1) + x(k)) + \frac{1}{2}(A+B)(x(k+1) - x(k)).$$

Here

$A - B = -n(k^2 + (n+2)k - n(n+1))$, $A + B = 2k^3 + (2-n)k^2 - (n^2 + 2n)k + (n^3 + n^2)$, so $\deg(A - B) = 2 < 3 = \deg(A + B)$ and one is in the second case of Gosper's degree bound. For x of degree m with leading coefficient x_m , the coefficient of k^{m+2} on the left is $\frac{1}{2}((-n) \cdot 2x_m + 2 \cdot mx_m) = x_m(m - n)$, which is nonzero in $\mathbb{Q}(n)$ because n is an indeterminate. Hence $\deg(Ax(k+1) - Bx(k)) = m + 2$, forcing $m = \deg c - 2 = 0$. With $x = x_0$ constant, the coefficient of k^2 in $x_0(A - B) = c(k)$ gives $x_0 = -1/n$; the coefficient of k is then $n + 2$, against the required $6n + 3$, and $n + 2 \neq 6n + 3$ in $\mathbb{Q}(n)$. Contradiction. [VERIFIED]: solving the linear system directly also returns no solution for any $\deg x \leq 8$. $\square$

Remark 6.13. The restriction to $\mathbb{Q}(n)$ is not cosmetic: for a *fixed* positive integer n the degeneracy $m = n$ is not excluded and $\gcd(a(k), b(k+n)) \neq 1$. The theorem rules out a certificate valid uniformly in n — the only kind that can prove a closed form — and says nothing about individual values of n .

Remark 6.14 (diagnosis). In the Franel and s_{10} cases the weight is $k \leftrightarrow n - k$ symmetric, the certificate module has rank 2 on $\{1, \delta\}$, and each component telescopes separately. For $\mathbf{D}$ the weight is genuinely three-lettered $(H_k^{(1)}, H_{n-k}^{(1)}, H_n^{(1)})$ and the components do not telescope separately, while the weight-two part of any enlarged ansatz is forced to vanish ($\Delta_k(Sz) = 0 \Rightarrow z = 0$). Closing $\mathbf{D}$ therefore needs either a different gauge for $\mathfrak{w}$ — the harmonic-monomial fit has a large kernel,

and a kernel shift changes the correction — or a genuine $\Pi\Sigma$ -field ansatz allowing nested sums, as in Schneider’s treatment of Apéry’s double sum [14]. [OPEN]. This is the first place in the programme where the rank-two mechanism is *provably* insufficient, and the obstruction is exactly the one met at weight 5: two independent letters at the top.

6.5. **E**: a twisted closed form.

Observation 6.15. [VERIFIED: $n \leq 50$ by direct re-implementation; $n \leq 20$ re-checked here; held out $n = 70, \dots, 82$ in the original fit] For **E**, $A(n) = \sum_k \binom{n}{k} \binom{2k}{k} \binom{2(n-k)}{n-k}$, the second solution is

$$B(n) = \sum_k \binom{n}{k} \binom{2k}{k} \binom{2(n-k)}{n-k} \left[\frac{1}{2} K_{\chi_{-4}}^{(2)}(2k) + \left(\frac{3}{4} H_k^{(1)} - \frac{1}{2} H_{2k}^{(1)} \right) (K_{\chi_{-4}}^{(1)}(2k) - K_{\chi_{-4}}^{(1)}(2n-2k)) \right],$$

with $B(0), B(1), B(2), B(3) = 0, 1, 7, \frac{404}{9}$, and $B(n)/A(n) \rightarrow G/2$, G Catalan’s constant.

This is the only known closed form for a sporadic second solution with a non-trivial character, and it is the empirical anchor of the whole χ -twist story: the same fitting problem is inconsistent in every pure harmonic basis and consistent as soon as χ_{-4} -letters are admitted — exactly what the heuristic of Remark 5.1 predicts for a limit of the form $L(\chi_{-4}, 2)/2$. Every argument of the weight is wide ($2k, 2n-2k$), and by the exhaustion recorded in §5.4 no tame alternative exists; this is why **E** sits outside Theorem 4.5.

7. RELATION TO BEUKERS–VLASENKO’S CONJECTURE 7.5

Set $f(x) = 1 - tg(x)$ for a Laurent polynomial g and $F(t) = \sum_n f_n t^n$ with $f_n = \text{ct}(g(x)^n)$; for the sporadic families these f_n are the $A(n)$ [10, 16]. Let $F_N(t)$ be the truncation of F at t^N , let $F(t) \log t + G(t)$ be the second solution of the Picard–Fuchs operator — so that the coefficients of G are our $B(n)$, up to normalisation — and let $q(t) = t \exp(G(t)/F(t))$ be the canonical (mirror) coordinate. Beukers–Vlasenko prove in [5, §7], building on [4], that for all $m, s \geq 1$,

$$\frac{F(t)}{F(t^\sigma)} \equiv \frac{F_{mp^s}(t)}{F_{mp^{s-1}}(t^\sigma)} \pmod{p^s}, \quad (7.1)$$

whose coefficientwise reading with the naive lift $t^\sigma = t^p$ and $s = 1$ is the first-row Lucas congruence (1.1). Their Conjecture 7.5 asserts (7.1) modulo p^{2s} for the *excellent* Frobenius lift — the unique lift with $q^\sigma = \gamma^{p-1} q^p$ — in the completely symmetric Calabi–Yau families; they state that they could not prove it for a single g .

The relationship to (LB_w^χ) is as follows.

- (1) **Two rows of one matrix.** The crystal $\text{CY}(g)(2)$ of [5] is free of rank 2 on $1/f, \theta(1/f)$, and its Frobenius has diagonal (λ_0, λ_2) with $\lambda_0 = F(t)/F(t^\sigma)$ the unit root. Conjecture 7.5 is a statement about λ_0 alone, sharpened from p^s to p^{2s} . Statement (LB_w^χ) is the statement about the *other* diagonal entry: that Frobenius acts on the second graded piece with eigenvalue $\chi(p)p^w$. In coefficient form, their λ_0 -congruence *is* the A -row and ours *is* the B -row. **Neither formally implies the other.**
- (2) **But 7.5’s standing hypothesis is a weak form of our conclusion.** Conjecture 7.5 is stated only for the excellent lift, which exists as a lift of $\mathbb{Z}_p[[t]]$ precisely because $q(t) = t \exp(G/F) \in \mathbb{Z}_p[[t]]$ — p -integrality of the mirror map, a statement about G , i.e. about $B(n)$. Observation 4.8 — $v_p(B(n)) \geq -w[\log_p n]$ — is exactly the coefficient-level shadow of that integrality, and it is sharper: it does not merely assert p -integrality, it pins the depth at exactly w , and (LB_w^χ) exhibits the eigenvalue $\chi(p)$ on top of it. Heuristically, then, $(\text{LB}_w^\chi) \rightsquigarrow$ coefficient form of mirror integrality $\implies$ the excellent lift is defined over $\mathbb{Z}_p[[t]]$,

the last being the standing hypothesis of 7.5. The first arrow is written with a wavy line deliberately: as Remark 4.9 explains, deducing Observation 4.8 from (LB_w^χ) needs both the

master form and a p -unit hypothesis on $A(n)$, so this is a statement about where the two circles of ideas sit relative to one another, not a chain of implications. In that sense (LB_w^χ) is *upstream* of 7.5, and certainly not equivalent to it.

- (3) **The character is invisible in unit-root formulations.** No character can appear in $\lambda_0 = F/F^\sigma$: the unit root of the holomorphic period is a unit, and the first-row Lucas congruence is untwisted for all fifteen families [11, 16]. The character surfaces only on the *non-unit* eigenvalue. Any crystal-theoretic proof of (LB_w^χ) must therefore carry the information that for seven of the fifteen families the second Frobenius eigenvalue is $\chi(p)p^w$ and not p^w , i.e. that the family is attached to a quadratic-twisted modular parametrisation. *This is a concrete correction to make to any “Dwork crystals $\Rightarrow$ Lucas congruences for the second solution” slogan.* We have not located a second-solution version of (7.1) in [5], nor elsewhere; we do not claim to have surveyed [3, 4] exhaustively.
- (4) **What (LB_w^χ) does not give.** Nothing towards the modulo- p^{2s} *doubling* that is the actual content of 7.5. Our floor is exactly w and equality is attained in every cell tested, so there is no hidden supercongruence in the B -row of the kind 7.5 predicts for the A -row.

Proposition 2.3 is the sharpest evidence for the Frobenius reading. For $\mathbf{B}$, δ and η the archimedean Apéry limit does not exist — the characteristic roots are complex conjugates of equal modulus and $B(n)/A(n)$ oscillates forever — yet (LB_w^χ) holds flat for all three, with $\chi = \chi_{-3}, \mathbf{1}, \chi_5$. The p -adic Apéry limit survives where the archimedean one does not; whatever (LB_w^χ) is measuring, it is not the limit.

8. MACHINE VERIFICATION

The following are formalised in Lean 4 with Mathlib and checked by the kernel with no `sorry` and no `native_decide`; for each of them

`#print axioms → [propext, Classical.choice, Quot.sound].`

declaration	statement
<code>theorem_LB</code>	Theorem 4.5 in the abstract form of Remark 4.7
<code>bMin_eq_bApery</code>	$\sum_k \binom{n}{k}^2 \binom{n+k}{k}^2 (2H_n^{(3)} - H_k^{(3)}) = b_n$, b_n defined by Apéry’s recurrence
<code>apery_b_harmonic_closed_form</code>	the same, with the harmonic sums written out
<code>bMin_lucas</code>	$p^3 \tilde{b}_{ap+r} \equiv \tilde{b}_a a_r \pmod{p}$ for the explicit sum, any prime p
<code>bApery_lucas</code>	the same for the sequence defined by Apéry’s recurrence
<code>bFanel_lucas</code>	$p^2 \beta_{ap+r} \equiv \beta_a f_r \pmod{p}$, $p \neq 2$
<code>apery_lucas</code>	Gessel’s congruence $a_{ap+r} \equiv a_a a_r \pmod{p}$, in digit-product form
<code>Q_lucas</code>	its analogue for a Brown–Zudilin double sum (not a sporadic first row)

Three scope remarks, stated because they are easy to overstate.

- (1) `bMin_lucas` carries *no* hypothesis $p \geq 5$: the minimal weight’s coefficients are the integers $2, -1$, so the (H4) coefficient clause is vacuous. The $p \geq 5$ of the informal statement is an artefact of Apéry’s original weight, which carries $1/(2m^3)$. Similarly `bFanel_lucas` needs only $p \neq 2$.
- (2) That `bFanel` is the second solution of the Fanel recurrence is [VERIFIED] upstream and *not* formalised; the Lean theorem is about the explicit sum. Theorem 6.5 of the present paper supplies the missing identification on paper, but it is not machine-checked. The corresponding identification *is* formalised in the γ case (`bMin_eq_bApery`).
- (3) The equivalence of the minimal form with van der Poorten’s classical double sum is proved on paper [26, §5] and is *not* formalised.

Nothing in §5 beyond Theorem 4.5 and its two instances is formalised; in particular Theorem 5.7, Theorem 6.12 and Corollary 6.9 are paper proofs.

A standing lesson of the programme belongs here: `#print axioms` cannot detect a false undischarged hypothesis. In a different part of the development a hypothesis that was false as stated left everything downstream “axiom-clean”; only an attempt to discharge it caught the error. The house rule that followed — no certificate reaches the kernel without a faithful re-implementation outside Lean — is why Remark 6.7 exists.

9. OPEN PROBLEMS

Ranked by the ratio of value to distance.

- P1. s_7 , two local bounds.** Problem 5.4: prove (V1) and (V2) for s_7 , $p \geq 5$, $p \neq 7$. Bounded, purely local, Kummer-versus-order-one-pole; both are [VERIFIED] with 0 violations in 1.66 million vanishing-layer terms and the minimum is exactly 1, so nothing is being asked for beyond what the data already shows. Proved count $4 \rightarrow 5$.
- P2. The descent congruence (LD).** Prove $V(n) \equiv \chi(p)^\varepsilon \Delta(a) A(r) \pmod{p}$ for $\alpha, \varepsilon, \mathbf{E}$. This is where the first instance with $\chi \neq 1$ is gated. Two leads: $\Delta(a) = 0$ for $a \leq (p-1)/2$, so it is a large-digit phenomenon; and the per-digit refinement is [VERIFIED] for ε and $\mathbf{E}$ (not for α), which makes those two provable one digit at a time. By Theorem 5.7 re-fitting the weight is excluded, so the proof must use something the fit does not see.
- P3. The master form.** Prove Conjecture 3.1 — the mod- p^w , all- n statement — in even one case. Theorem 4.5 gives only the single-digit mod- p shadow. The gap is the same one left open at weight 3 in [21].
- P4. New letters for the conjectural seven.** By the rank diagnostic of §5.5, $\mathbf{B}, \delta, \zeta$ admit no $\mathbb{Q}$ -relation at all between summand and harmonic letters, so no enlargement of a $\sum_k S \cdot \mathbf{w}$ fit will succeed. The two candidates are (a) Gorodetsky-style constant-term representations [10], now known to be necessary rather than optional for $\mathbf{C}$ (Proposition 5.10), and (b) rational-function coefficients $c_j(n, k)$, which would require replacing (H4)’s coefficient clause by $c_j(n, k) \equiv c_j(a, b) \pmod{p}$ on the surviving set — note that the obvious candidates $1/(2k+1)$ and $1/(n+1)$ fail that test.
- P5. D’s closed form.** Upgrade Observation 6.11 to a theorem. Theorem 6.12 says the rank-two mechanism cannot do it; try a gauge change inside the 11-dimensional kernel of the fit, or a $\Pi\Sigma$ ansatz [14]. This is a miniature of the weight-5 blockage in a case small enough to compute, and it should be settled before anyone spends weeks at weight 5.
- P6. Deformations as a source of letters.** Straub–Zudilin’s deformation $A^{(s)}(n, t)$ (Remark 6.8) produces the entire harmonic weight of $\sum_k \binom{n}{k}^s$ as one Taylor coefficient. Does an analogue exist for the other sporadic summands? This is a strictly larger class than the one already excluded in §5.5: the programme’s earlier “ Γ -derivation” probe covered only *degree-one* weights $\sum_x \lambda_x H_x^{(w)}$ and found γ alone consistent, whereas the deformation above yields the degree-two weight $\alpha\sigma + \beta\delta^2$. Since $\mathbf{B}, \delta$ and ζ admit no $\mathbb{Q}$ -relation at all between summand and harmonic letters, a deformation — which changes the summand, not the alphabet — is precisely the kind of move that could supply letters no fit can find.
- P7. A crystal-theoretic proof.** Give the argument of §7 its natural home: a Frobenius-eigenvalue statement on the second graded piece of $\text{CY}(g)(2)$, carrying the character. The three families with no archimedean limit are the test cases, because for them no archimedean argument can be smuggled in.

PROVENANCE AND DATA

Every statement above is traceable to one of the following memos of the programme, which contain the scripts and the raw logs: [23] (the fifteen pairs, the sweeps, the decompositions, Lemma K and Theorem LB), [24] (the (H2) correction, Propositions 5.5 and 5.6, Theorem 5.7),

[25] (the complete tame searches, Propositions 5.10 and 5.11, the rank diagnostic), [26] (§6). All arithmetic underlying a [VERIFIED] label is exact — **Fraction** arithmetic or fixed-precision p -adics cross-validated against it cell by cell — and no verdict rests on a single modulus: every negative in §5.5 and §5.4 was recomputed at a second, unrelated prime with identical rank and verdict. For the present paper the following were re-derived independently: the recurrences of all fifteen binomial sums ($n \leq 12$); the five decompositions of §5.1 ($n \leq 20$); the twisted congruence (1.2) for all fifteen families at $p = 5, 7, 11, 13$ (4920 cells, 0 failures); the master floor for all fifteen at $p = 5, 7$ and all $n < p^3$ (exactly w in all 30 cells); the discriminants of Proposition 2.3; the weights of Observation 2.2 ($n \leq 120$); and every certificate identity of §6.2–§6.4.

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